



A nonlocal in time parabolic system whose Fujita critical exponent is not given by scaling

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ABSTRACT

We consider the nonlocal coupled parabolic system

$$u_t - \Delta u = \int_0^t (t-s)^{-\gamma_1} |v|^{p-1} v(s) ds, \quad v_t - \Delta v = \int_0^t (t-s)^{-\gamma_2} |u|^{q-1} u(s) ds$$

in $(0, T) \times \Omega$ when either $\Omega = \mathbb{R}^N$ or $\Omega \subset \mathbb{R}^N$ is a bounded domain with smooth boundary. We assume that $0 \leq \gamma_1, \gamma_2 < 1$, $p, q \geq 1$ and the initial data $u(0), v(0) \in C_0(\Omega)$. We obtain the Fujita critical exponent for the system above which is not given by the scaling argument.

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1. Introduction

We consider the following semilinear parabolic system with a nonlocal in time nonlinearity

$$\begin{cases} u_t - \Delta u = \int_0^t (t-s)^{-\gamma_1} |v|^{p-1} v(s) ds & \text{in } (0, T) \times \mathbb{R}^N, \\ v_t - \Delta v = \int_0^t (t-s)^{-\gamma_2} |u|^{q-1} u(s) ds & \text{in } (0, T) \times \mathbb{R}^N, \\ u(0) = u_0, \quad v(0) = v_0 & \text{in } \mathbb{R}^N, \end{cases} \quad (1)$$

with $p, q \geq 1$, $0 \leq \gamma_1, \gamma_2 < 1$ and initial data $u_0, v_0 \in C_0(\mathbb{R}^N)$. The space $C_0(\mathbb{R}^N)$ denotes the set of all continuous functions decaying to zero at infinity.

As usual, the system (1) is equivalent, in a appropriate setting, to the integral system

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$$\begin{cases} u(t) = S(t)u_0 + \int_0^t \int_0^s (s-\sigma)^{-\gamma_1} S(t-s)|v|^{p-1}v(\sigma) d\sigma ds, \\ v(t) = S(t)v_0 + \int_0^t \int_0^s (s-\sigma)^{-\gamma_2} S(t-s)|u|^{q-1}u(\sigma) d\sigma ds \end{cases} \quad (2)$$

for all $t \in [0, T]$, where $\{S(t)\}_{t \geq 0}$ is the heat semigroup on $\mathbb{R}^N$.

Our first result concerns the existence and uniqueness of solutions for the system (1).

Theorem 1. Consider $p, q \geq 1$, $\gamma_1, \gamma_2 \in [0, 1)$ and $u_0, v_0 \in C_0(\mathbb{R}^N)$. There exists a unique function $(u, v) \in \{C([0, T_{\max}), C_0(\mathbb{R}^N))\}^2$ solution of (1) such that either

- (i) $T_{\max} = \infty$ (the solution is global) or else
- (ii) $T_{\max} < \infty$ and $\lim_{t \rightarrow T_{\max}} (\|u(t)\|_{\infty} + \|v(t)\|_{\infty}) = \infty$ (the solution blows up in finite time).

Furthermore, if $(u_0, v_0) \neq (0, 0)$, $u_0, v_0 \geq 0$, then $u(t), v(t) \geq 0$ for all $t \in (0, T_{\max})$. For all $0 < t \leq t + \tau < T_{\max}$, one has

$$u(t + \tau) \geq S(\tau)u(t), \quad v(t + \tau) \geq S(\tau)v(t). \quad (3)$$

Moreover, if $(u_0, v_0) \in L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N)$ with $1 \leq r_2 \leq pr_1$ and $1 \leq r_1 \leq qr_2$, then $(u, v) \in C([0, T_{\max}), L^{r_1}(\mathbb{R}^N)) \times C([0, T_{\max}), L^{r_2}(\mathbb{R}^N))$, and

$$\lim_{t \rightarrow T_{\max}} (\|u(t)\|_{\infty} + \|v(t)\|_{\infty} + \|u(t)\|_{r_1} + \|v(t)\|_{r_2}) = \infty, \quad (4)$$

when $T_{\max} < \infty$.

We are also interested in finding conditions that determine when solutions of system (1) either blow up in finite time or exist globally.

In order to motivate it, it is convenient to recall some facts for the semilinear parabolic problem

$$\begin{cases} u_t - \Delta u = \int_0^t (t-s)^{-\gamma} |u|^{p-1}u(s) ds & \text{in } (0, T) \times \mathbb{R}^N, \\ u(0) = u_0 & \text{in } \mathbb{R}^N \end{cases} \quad (5)$$

with $p > 1$, $0 \leq \gamma < 1$ and $u_0 \in C_0(\mathbb{R}^N)$. Problem (5) was studied in [2]. There, the authors showed that if $u \in C([0, T_{\max}), C_0(\mathbb{R}^N))$ is the solution of (5) and $p_* = \max\{\frac{1}{\gamma}, p_{\gamma}\}$ ($p_* = +\infty$ if $\gamma = 0$), where

$$p_{\gamma} = \begin{cases} 1 + \frac{4-2\gamma}{N-2+2\gamma}, & N \geq 2 \text{ or } \left(N = 1 \text{ and } \frac{1}{2} < \gamma < 1\right), \\ +\infty, & N = 1 \text{ and } 0 < \gamma \leq \frac{1}{2}, \end{cases} \quad (6)$$

then

- (i) If $p \leq p_*$ and $u_0 \geq 0$, $u_0 \not\equiv 0$, then u blows up in finite time.
- (ii) If $p > p_*$ and $u_0 \in L^{q_{sc}}(\mathbb{R}^N)$, where $q_{sc} = \frac{N(p-1)}{4-2\gamma}$, with $\|u_0\|_{q_{sc}}$ sufficiently small, then u exists globally.

Since p_* is a critical value for the existence of global solutions, it is called the Fujita critical exponent of problem (5). This term has its origin in the study of the semilinear parabolic equation

$$u_t - \Delta u = |u|^{p-1}u \quad \text{in } (0, T) \times \mathbb{R}^N \quad (7)$$

carried out in [6,7,9,10]. In this case, the Fujita critical exponent is $p_* = 1 + \frac{2}{N}$ and can be determined using the following argument: if $u(t, x)$ is a solution of Eq. (7) with initial data u_0 , then $\lambda^{\frac{2}{p-1}}u(\lambda^2t, \lambda x)$ is also a solution of (7) with initial data $\lambda^{\frac{2}{p-1}}u_0(\lambda x)$. We have

$$\|\lambda^{\frac{2}{p-1}}u_0(\lambda \cdot)\|_r = \lambda^{\frac{2}{p-1} - \frac{N}{r}} \|u_0\|_r. \quad (8)$$

From here, we deduce that the norm in L^r is invariant if and only if $r = r_c = \frac{N(p-1)}{2}$. Moreover, $r_c > 1$ if and only if $p > p_*$. Note that $L^{r_c}(\mathbb{R}^N)$ is the only space where small initial data imply the existence of global solutions. Indeed, if we assume

that $r \neq r_c$ and that the norm $\|u_0\|_r$ sufficiently small assures that the solution is global, then we deduce from (8) that all solution of (7) with initial data $u_0 \in L^r(\mathbb{R}^N)$ is global. This is obviously false since the solution of (7) blows up in finite time if the initial data is large enough.

The previous argument is called scaling argument. It can be used to find the Fujita critical exponent of other equations as, for example, $u_t = \Delta u + t^k |x|^\sigma u^p$ (see [1,11–13]), $u_t = \nabla \cdot (u^\sigma \nabla u) + u^p$ (see [8]) and $u_t = \Delta u + u(t, 0)^{p-q} u(t, x)^q$, with $p > q \geq 1$ (see [14]). However, in [2], the authors showed that the Fujita critical exponent p_* of Eq. (5) is not given by this argument. In fact, if $u(t, x)$ is solution of (5) with initial data $u_0 \in L^r(\mathbb{R}^N)$, then $\lambda^{\frac{4-2\gamma}{p-1}} u(\lambda^2 t, \lambda x)$ is also a solution of (5) with initial data $\lambda^{\frac{4-2\gamma}{p-1}} u_0(\lambda x)$. Since $\|\lambda^{\frac{4-2\gamma}{p-1}} u_0(\lambda \cdot)\|_r = \lambda^{\frac{4-2\gamma}{p-1} - \frac{N}{r}} \|u_0\|_r$, the norm in $L^r(\mathbb{R}^N)$ is preserved if and only if $r = r_{sc} = \frac{N(p-1)}{4-2\gamma}$. Note that $r_{sc} > 1$ if and only if $p > p_{sc} = 1 + \frac{4-2\gamma}{N}$. However, from [2], $p_{sc} < p_*$.

The scaling argument can also be used to find the Fujita critical exponent of some systems. For example, if (u, v) is a solution of the system

$$\begin{cases} u_t - \Delta u = |v|^{p-1} v & \text{in } (0, T) \times \mathbb{R}^N, \\ v_t - \Delta v = |u|^{q-1} u & \text{in } (0, T) \times \mathbb{R}^N, \\ u(0) = u_0, \quad v(0) = v_0 & \text{in } \mathbb{R}^N, \end{cases} \quad (9)$$

with $p, q > 0$ and $pq > 1$, then (u_λ, v_λ) defined by

$$u_\lambda(t, x) = \lambda^\gamma u(\lambda^2 t, \lambda x), \quad v_\lambda(t, x) = \lambda^\delta v(\lambda^2 t, \lambda x) \quad (10)$$

where $\gamma = \frac{2(q+1)}{pq-1}$, $\delta = \frac{2(p+1)}{pq-1}$, is also a solution of this system with initial data $(\lambda^{\frac{2(q+1)}{pq-1}} u_0(\lambda x), \lambda^{\frac{2(p+1)}{pq-1}} v_0(\lambda x))$. Therefore, we have $\|u_\lambda(0)\|_r = \lambda^{\frac{2(q+1)}{pq-1} - \frac{N}{r}} \|u_0\|_r$, $\|v_\lambda(0)\|_s = \lambda^{\frac{2(p+1)}{pq-1} - \frac{N}{s}} \|v_0\|_s$, and these norms are invariant when $r_c = \frac{N}{2}(\frac{pq-1}{q+1})$ and $s_c = \frac{N}{2}(\frac{pq-1}{p+1})$. Note that $r_c > 1$, $s_c > 1$ if and only if $\frac{q+1}{pq-1} < \frac{2}{N}$ and $\frac{p+1}{pq-1} < \frac{2}{N}$, which coincide with the results obtained in [4]: if $p, q > 0$ with $pq > 1$, then

- (i) If $\frac{p+1}{pq-1} < \frac{2}{N}$, $\frac{q+1}{pq-1} < \frac{2}{N}$ and $(u_0, v_0) \in L^{r_c}(\mathbb{R}^N) \times L^{s_c}(\mathbb{R}^N)$ with $\|u_0\|_{r_c} + \|v_0\|_{s_c}$ is sufficiently small, then the solution of (9) exists globally.
- (ii) If $\frac{p+1}{pq-1} \geq \frac{2}{N}$ or $\frac{q+1}{pq-1} \geq \frac{2}{N}$ and $(u_0, v_0) \neq (0, 0)$, $u_0, v_0 \geq 0$ then the solution of (9) blows up in finite time.

For the system (1), if (u, v) is a solution with initial data (u_0, v_0) , then for all $\lambda > 0$, (u_λ, v_λ) defined by $u_\lambda(t, x) = \lambda^\gamma u(\lambda^2 t, \lambda x)$, $v_\lambda(t, x) = \lambda^\delta v(\lambda^2 t, \lambda x)$, where $\gamma = \frac{2\gamma_1 - 4 + p(2\gamma_2 - 4)}{1 - pq}$ and $\delta = \frac{2\gamma_2 - 4 + q(2\gamma_1 - 4)}{1 - pq}$, is also a solution of (1). If $(u_0, v_0) \in L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N)$, then the norms in $L^{r_1}(\mathbb{R}^N)$ and $L^{r_2}(\mathbb{R}^N)$ are preserved if and only if

$$r_1 = \frac{N(pq-1)}{2[2-\gamma_1+p(2-\gamma_2)]}, \quad r_2 = \frac{N(pq-1)}{2[2-\gamma_2+q(2-\gamma_1)]}. \quad (11)$$

Moreover, $r_1 > 1$ and $r_2 > 1$ if and only if

$$p(2-\gamma_2) + (2-\gamma_1) < \frac{N(pq-1)}{2} \quad \text{and} \quad q(2-\gamma_1) + (2-\gamma_2) < \frac{N(pq-1)}{2}. \quad (12)$$

We could expect that if (12) holds and $\|u_0\|_{r_1} + \|v_0\|_{r_2}$ is sufficiently small, with r_1 and r_2 given by (11), then the solution of (1) would exist globally. The following result shows partially that this is not true.

Theorem 2. Let $p, q \geq 1$, $pq > 1$, $0 \leq \gamma_1, \gamma_2 < 1$ and $u_0, v_0 \in C_0(\mathbb{R}^N)$. Let $(u, v) \in \{C([0, T_{\max}), C_0(\mathbb{R}^N))\}^2$ be the correspondent solution of (1).

(i) Assume that

$$\begin{cases} 1 - p\gamma_2 + p(1 - q\gamma_1) + p(q+1) < \frac{N}{2}(pq-1), \\ 1 - q\gamma_1 + q(1 - p\gamma_2) + q(p+1) < \frac{N}{2}(pq-1), \end{cases} \quad (13)$$

$$\begin{cases} 1 - p\gamma_2 + p(1 - q\gamma_1) < 0, \\ 1 - q\gamma_1 + q(1 - p\gamma_2) < 0 \end{cases} \quad (14)$$

and

$$\frac{p}{r_2} - \frac{4}{N} < \frac{1}{p}, \quad \frac{q}{r_1} - \frac{4}{N} < \frac{1}{q}. \quad (15)$$

If $(u_0, v_0) \in L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N)$, where r_1, r_2 are defined in (11) and $(\|u_0\|_\infty + \|v_0\|_\infty + \|u_0\|_{r_1} + \|v_0\|_{r_2}) \leq \epsilon$ with $\epsilon > 0$ sufficiently small, then (u, v) exists globally.

(ii) Assume that

$$\begin{cases} 1 - p\gamma_2 + p(1 - q\gamma_1) + p(q + 1) \geq \frac{N}{2}(pq - 1) \\ \text{or} \\ 1 - q\gamma_1 + q(1 - p\gamma_2) + q(p + 1) \geq \frac{N}{2}(pq - 1) \end{cases} \quad (16)$$

or

$$\begin{cases} 1 - p\gamma_2 + p(1 - q\gamma_1) \geq 0 \\ \text{or} \\ 1 - q\gamma_1 + q(1 - p\gamma_2) \geq 0. \end{cases} \quad (17)$$

If $(u_0, v_0) \neq (0, 0)$ with $u_0, v_0 \geq 0$, then (u, v) blows up in finite time.

Remark 1. Here are some comments about Theorem 2.

- (i) In the case $p = q$, $\gamma_1 = \gamma_2 = \gamma$, the condition (15) is satisfied if the condition (14) is valid. Indeed, in this case we have $r_1 = r_2 = r = \frac{N(p-1)}{2(2-\gamma)}$ and $\frac{p}{r} - \frac{2}{N} < 1$. Thus, we have $\frac{p}{r} - \frac{4}{N} < 1 - \frac{2}{N}$. If $1 - \frac{2}{N} \leq \frac{1}{p}$, then $\frac{p}{r} - \frac{4}{N} < \frac{1}{p}$. Suppose that $\frac{1}{p} < 1 - \frac{2}{N}$ that is $p < \frac{N(p-1)}{2}$. By the condition (14), follows that $p\gamma > 1$ and we have $2p - p^2\gamma < p < \frac{N(p-1)}{2}$. Hence, $\frac{p}{r} - \frac{4}{N} = \frac{2(2-p\gamma)}{N(p-1)} < \frac{1}{p}$.
- (ii) If the conditions (13) and (14) are valid, then $\frac{p}{r_2} - \frac{4}{N} \geq \frac{1}{q}$ and $\frac{q}{r_1} - \frac{4}{N} \geq \frac{1}{p}$ does not occur. Indeed, by (13) follows that $\frac{1}{p} > \frac{N}{2}(\frac{1}{r_2} - \frac{1}{p})$. Moreover, the conditions $\frac{p}{r_2} - \frac{4}{N} \geq \frac{1}{q}$ and $\frac{q}{r_1} - \frac{4}{N} \geq \frac{1}{p}$ imply $1 > p\gamma_2$ and $1 > q\gamma_1$ respectively. This fact contradicts the condition (14).
- (iii) Theorem 2 is sharp in the case (15), see (13)–(17), and generalizes Theorem 1.1 of [2]. Indeed, let $p = q$ and $\gamma_1 = \gamma_2 = \gamma$. Conditions (13) and (14) reduce to $p(\frac{N}{2} + \gamma - 1) > \frac{N}{2} + 1$ and $p\gamma > 1$ respectively. Moreover, conditions (16) and (17) reduce to $p(\frac{N}{2} + \gamma - 1) \leq \frac{N}{2} + 1$ or $p\gamma \leq 1$. From these facts, it is possible to conclude that, if (15) is valid, then the value of the Fujita critical exponent is p_* .
- (iv) It is interesting to observe that when $\gamma_1 = 0$ or $\gamma_2 = 0$ all nontrivial positive solutions blow up in finite time. Indeed, suppose without loss of generality that $\gamma_1 = 0$. Since $0 \leq \gamma_2 < 1$ we see that $1 - p\gamma_2 + p(1 - q\gamma_1) = 1 + p(1 - \gamma_2) > 0$. Thus, the first inequality of (17) is satisfied.
- (v) Recently, using techniques different from the ones used here, Fino and Kirane [5] have considered the system (1) and have shown assuming $u_0, v_0 \in C_0(\mathbb{R}^N) \cap L^2(\mathbb{R}^N)$, $(u_0, v_0) \neq (0, 0)$, $u_0, v_0 \geq 0$ that if (16) or $(p\gamma_2 < 1 \text{ and } q\gamma_1 < 1)$ hold then any solution of (1) blows up in finite time. Clearly, the conditions used by them imply (16) or (17).

For the system (1) in a bounded domain $\Omega \subset \mathbb{R}^N$ with Dirichlet boundary conditions

$$\begin{cases} u_t - \Delta u = \int_0^t (t-s)^{-\gamma_1} |v|^{p-1} v(s) ds & \text{in } (0, T) \times \Omega, \\ v_t - \Delta v = \int_0^t (t-s)^{-\gamma_2} |u|^{q-1} u(s) ds & \text{in } (0, T) \times \Omega, \\ u(t, x) = 0, \quad v(t, x) = 0 & \text{in } (0, T) \times \partial\Omega, \\ u(0) = u_0, \quad v(0) = v_0 & \text{in } \Omega \end{cases} \quad (18)$$

with $p, q \geq 1$, $pq > 1$, $0 \leq \gamma_1, \gamma_2 < 1$ and initial data $u_0, v_0 \in C_0(\mathbb{R}^N)$, we have the following result.

Theorem 3. Let Ω be a smooth, bounded domain of $\mathbb{R}^N$, $p, q \geq 1$, $pq > 1$, $0 \leq \gamma_1, \gamma_2 < 1$ and $u_0, v_0 \in C_0(\Omega)$. Let $(u, v) \in \{C[0, T_{\max}), C_0(\Omega)\}^2$ be the correspondent solution of (18).

(i) Assume that

$$\begin{cases} 1 - p\gamma_2 + p(1 - q\gamma_1) < 0 \\ \text{and} \\ 1 - q\gamma_1 + q(1 - p\gamma_2) < 0. \end{cases} \quad (19)$$

If $\|u_0\|_\infty \leq \delta_1$ and $\|v_0\|_\infty \leq \delta_2$, where $\delta_1, \delta_2 > 0$ are sufficiently small, then (u, v) exists globally.

(ii) Assume that

$$\begin{cases} 1 - p\gamma_2 + p(1 - q\gamma_1) \geq 0 \\ \text{or} \\ 1 - q\gamma_1 + q(1 - p\gamma_2) \geq 0. \end{cases} \quad (20)$$

If $(u_0, v_0) \neq (0, 0)$ with $u_0, v_0 \geq 0$, then (u, v) blows up in finite time.

Remark 2.

- (i) An existence and uniqueness result is also valid for the system (18). Its proof is similar to the proof of Theorem 1.
- (ii) Theorem 3 is sharp and extends to system (1) Theorem 1.3 of [2].
- (iii) If $\gamma_1 = 0$ or $\gamma_2 = 0$ all nontrivial positive solutions blow up in finite time (see (iii) of Remark 1). This fact, for Eq. (5), was first observed by Souplet [15].

The rest this paper is organized as follows. In the next section we show Theorem 1. Sections 3 and 4 are devoted to show Theorem 2. Finally, in Section 5, Theorem 3 is proved.

2. Proof of the Theorem 1

Lemma 4. Let $T > 0$, $A, B \geq 0$, and let f_1, f_2 be nonnegative functions with $f_1, f_2 \in L^1(0, T)$. Consider two nonnegative and nondecreasing functions $\varphi_1, \varphi_2 \in C[0, T]$ such that

$$\begin{cases} \varphi_1(t) \leq A + \int_0^t f_1(s)\varphi_2(s) ds, \\ \varphi_2(t) \leq B + \int_0^t f_2(s)\varphi_1(s) ds. \end{cases} \quad (21)$$

Then there exists a constant $C > 0$ such that

$$\varphi_1(t) \leq C(A + B) \exp\left(\int_0^t C f_1(s) ds\right) \quad \text{and} \quad \varphi_2(t) \leq C(A + B) \exp\left(\int_0^t C f_2(s) ds\right),$$

for almost all $t \in [0, T]$.

Proof. The result follows easily, replacing one of the inequalities in (21) by the other one and use the Gronwall's inequality. $\square$

Proof of Theorem 1. Let $M \geq \|u_0\|_\infty + \|v_0\|_\infty$. Consider the set

$$K = \{(u, v) \in \{L^\infty((0, T), C_0(\mathbb{R}^N))\}^2; \|u(t)\|_\infty, \|v(t)\|_\infty \leq M + 1\}, \quad (22)$$

where $T > 0$ will be chosen later. The space K endowed with the metric

$$d((u, v), (\bar{u}, \bar{v})) = \sup_{t \in (0, T)} \|u(t) - \bar{u}(t)\|_\infty + \sup_{t \in (0, T)} \|v(t) - \bar{v}(t)\|_\infty$$

is a complete metric space. Consider the mapping $\psi : K \rightarrow E$ defined by $\psi(u, v) = (\varphi_1(v), \varphi_2(u))$, where

$$\begin{cases} \varphi_1(v)(t) = S(t)u_0 + \int_0^t \int_0^s (s - \sigma)^{-\gamma_1} S(t - s) |v|^{p-1} v(\sigma) d\sigma ds, \\ \varphi_2(u)(t) = S(t)v_0 + \int_0^t \int_0^s (s - \sigma)^{-\gamma_2} S(t - s) |u|^{q-1} u(\sigma) d\sigma ds. \end{cases} \quad (23)$$

For $(u, v) \in K$, we have

$$\begin{aligned} \|\varphi_1(v)(t)\|_\infty &\leq \|u_0\|_\infty + \int_0^t \int_0^s (s - \sigma)^{-\gamma_1} \|v^p(\sigma)\|_\infty d\sigma ds \\ &\leq \|u_0\|_\infty + CT^{2-\gamma_1} (M + 1)^p. \end{aligned} \quad (24)$$

Just as before, we obtain

$$\|\varphi_2(u)(t)\|_\infty \leq \|v_0\|_\infty + T^{2-\gamma_2} C(M+1)^q. \quad (25)$$

Arguing as in (24), it is possible to conclude

$$\|\varphi_1(v) - \varphi_1(\bar{v})\|_\infty \leq 2C(M+1)^{p-1} T^{2-\gamma_1} \sup_{t \in (0, T)} \|v(t) - \bar{v}(t)\|_\infty, \quad (26)$$

$$\|\varphi_2(u) - \varphi_2(\bar{u})\|_\infty \leq 2C(M+1)^{q-1} T^{2-\gamma_2} \sup_{t \in (0, T)} \|u(t) - \bar{u}(t)\|_\infty, \quad (27)$$

for $(u, v), (\bar{u}, \bar{v}) \in K$ and some constant $C > 0$.

From (24)–(27) follow that if T is sufficiently small, then ψ is a strict contraction from K into itself, so ψ has a unique fixed point (u, v) in K , which is a solution of (1).

Uniqueness follows from Lemma 4. Due to uniqueness, it follows that the solution (u, v) can be extended to a maximal interval $[0, T_{\max})$. Note that, if $0 \leq t \leq t + \tau \leq T_{\max}$, we have

$$\begin{aligned} u(t + \tau) &= S(\tau)u(t) + \int_0^\tau S(\tau - s) \int_0^s (s - \sigma)^{-\gamma_1} |v|^{p-1} v(t + \sigma) d\sigma ds \\ &\quad + \int_0^\tau S(\tau - s) \int_0^t (t + s - \sigma)^{-\gamma_1} |v|^{p-1} v(\sigma) d\sigma ds, \\ v(t + \tau) &= S(\tau)v(t) + \int_0^\tau S(\tau - s) \int_0^s (s - \sigma)^{-\gamma_2} |u|^{q-1} u(t + \sigma) d\sigma ds \\ &\quad + \int_0^\tau S(\tau - s) \int_0^t (t + s - \sigma)^{-\gamma_2} |u|^{q-1} u(\sigma) d\sigma ds. \end{aligned} \quad (28)$$

By a fixed point argument, it follows from (28) that if $\|u\|_{L^\infty((0, T) \times \mathbb{R}^N)} + \|v\|_{L^\infty((0, T) \times \mathbb{R}^N)} < \infty$, then (u, v) can be extended to an interval $[0, T')$ with $T' > T$. This shows that if $T_{\max} < \infty$, then $\lim_{t \rightarrow T_{\max}} (\|u(t)\|_\infty + \|v(t)\|_\infty) = \infty$.

If $u_0, v_0 \geq 0$, we construct a nonnegative solution using a fixed point argument in the set $K_+ = \{(u, v) \in K; u(t), v(t) \geq 0 \text{ for all } t \in (0, T)\}$, where K is given by (22). From (28), we have that u and v are nonnegative on $[0, T_{\max})$. Using this fact and (28) we get (3).

To show the remaining part of Theorem 1, we use again a fixed point argument. Consider the spaces

$$E = L^\infty((0, T), L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N) \cap \{L^\infty(\mathbb{R}^N)\}^2)$$

and $K = \{\bar{u} = (u, v) \in E; \|u(t)\|_\infty, \|v(t)\|_\infty, \|u(t)\|_{r_1}, \|v(t)\|_{r_2} < M + 1, \text{ for all } t \in (0, T)\}$ where $M \geq \max\{\|u_0\|_\infty, \|u_0\|_{r_1}, \|v_0\|_\infty, \|v_0\|_{r_2}\}$. Since $r_2 \leq pr_1$ and $r_1 \leq qr_2$ we can choose $\eta, \omega \geq 1$ satisfying

$$\begin{aligned} \frac{1}{pr_1} &\leq \frac{1}{\omega} \leq \min\left\{\frac{1}{p}, \frac{1}{r_2}, \frac{1}{p}\left(\frac{2}{N} + \frac{1}{r_1}\right)\right\}, \\ \frac{1}{qr_2} &\leq \frac{1}{\eta} \leq \min\left\{\frac{1}{q}, \frac{1}{r_1}, \frac{1}{q}\left(\frac{2}{N} + \frac{1}{r_2}\right)\right\}. \end{aligned}$$

The space (K, d) with the metric

$$d(\bar{u}, \bar{v}) = \max_{i=1,2} \left\{ \sup_{t \in (0, T)} \|u_i(t) - v_i(t)\|_\infty, \sup_{t \in (0, T)} \|u_i(t) - v_i(t)\|_{r_i} \right\},$$

where $\bar{u} = (u_1, u_2)$, $\bar{v} = (v_1, v_2)$ is a complete metric space.

Let $\psi : K \rightarrow E$ be the mapping given by (23). Let $\bar{w} = (u, v)$ and $\bar{z} = (\bar{u}, \bar{v})$ be in K . Proceeding as in (24) we obtain

$$\|\varphi_1(v)\|_\infty \leq M + CT^{2-\gamma_1}(M+1)^p, \quad (29)$$

$$\|\varphi_1(v) - \varphi_1(\bar{v})\|_\infty \leq CT^{2-\gamma_1}(M+1)^{p-1} \sup_{t \in (0, T)} \|v(t) - \bar{v}(t)\|_\infty. \quad (30)$$

Since $\omega \geq r_2$, from the interpolation inequality, we get

$$\begin{aligned}
\|\varphi_1(v)(t)\|_{r_1} &\leq \|u_0\|_{r_1} + \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{\omega}-\frac{1}{r_1})} \int_0^s (s-\sigma)^{-\gamma_1} \|v(\sigma)\|_{\omega}^p d\sigma ds \\
&\leq \|u_0\|_{r_1} + \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{\omega}-\frac{1}{r_1})} \int_0^s (s-\sigma)^{-\gamma_1} \\
&\quad \cdot \|v\|_{r_2}^{\frac{r_2}{\omega}p} \|v\|_{\infty}^{(1-\frac{r_2}{\omega})p} d\sigma ds \\
&\leq M + CT^{2-\gamma_1-\frac{N}{2}(\frac{p}{\omega}-\frac{1}{r_1})}(M+1)^p, \tag{31} \\
\|\varphi_1(v)(t) - \varphi_1(\bar{v})(t)\|_{r_1} &\leq \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{\omega}-\frac{1}{r_1})} \int_0^s (s-\sigma)^{-\gamma_1} \\
&\quad \cdot (\|v\|_w^{p-1} + \|\bar{v}\|_w^{p-1}) \|v(\sigma) - \bar{v}(\sigma)\|_{\omega} d\sigma ds \\
&\leq 2(M+1)^{p-1} d(\bar{w}, \bar{z}) \\
&\quad \cdot \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{\omega}-\frac{1}{r_1})} \int_0^s (s-\sigma)^{-\gamma_1} d\sigma ds \\
&\leq 2C(M+1)^{p-1} T^{2-\gamma_1-\frac{N}{2}(\frac{p}{\omega}-\frac{1}{r_1})} d(\bar{w}, \bar{z}). \tag{32}
\end{aligned}$$

Similarly, by analogous computations one can prove that

$$\|\varphi_2(u)\|_{\infty} \leq M + T^{2-\gamma_2}(M+1)^q, \tag{33}$$

$$\|\varphi_2(u) - \varphi_2(\bar{u})\|_{\infty} \leq T^{2-\gamma_2}(M+1)^{q-1} \sup_{t \in (0, T)} \|u(t) - \bar{u}(t)\|_{\infty}, \tag{34}$$

$$\|\varphi_2(u)(t)\|_{r_2} \leq M + T^{2-\gamma_2-\frac{N}{2}(\frac{q}{\eta}-\frac{1}{r_2})}(M+1)^q, \tag{35}$$

$$\|\varphi_2(u)(t) - \varphi_2(\bar{u})(t)\|_{r_2} \leq 2(M+1)^{q-1} T^{2-\gamma_2-\frac{N}{2}(\frac{q}{\eta}-\frac{1}{r_2})} d(\bar{w}, \bar{z}). \tag{36}$$

From (29), (31), (33), and (35), $\psi(K) \subset K$ for T small enough. From (30), (32), (34), and (36), $\psi : K \rightarrow K$ is a strict contraction for a possibly T smaller. It follows that ψ has a unique fixed point $\bar{u} = (u, v)$ in K .

Since $\bar{u} = (u, v) \in K$, we conclude that $\bar{u} = (u, v) \in C([0, T], L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N))$ for T small. The persistence in $L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N)$ after T is easily proved. The limit (4) follows arguing as in the previous case. $\square$

3. Proof of the global existence part of Theorem 2

We use the following smoothing effect of the heat semigroup: for $u_0 \in L^r(\mathbb{R}^N) \cap L^s(\mathbb{R}^N)$, $1 \leq r \leq s$, there exists a constant $C = C(r, s)$ such that

$$\|S(t)u_0\|_{L^s} \leq C(t+1)^{-\frac{N}{2}(\frac{1}{r}-\frac{1}{s})} (\|u_0\|_r + \|u\|_s) \tag{37}$$

for $t \geq 0$. We also use the following estimates

$$\int_0^t (t-s)^{-a} (1+s)^{-b} ds \leq C(1+t)^{1-a-b} \tag{38}$$

for $t > 0$, $0 < a, b < 1$, and

$$\int_0^t (t+1-s)^{-a} (1+s)^{-b} ds \leq C(1+t)^{-b} \tag{39}$$

for $t > 0$, $a > 1$, $a \geq b$ (see Lemma 4.1 of [3]). Here, C is a constant depending only on a and b .

Let $(u_0, v_0) \in \{C_0(\mathbb{R}^N)\}^2 \cap L^{r_1}(\mathbb{R}^N) \times L^{r_2}(\mathbb{R}^N)$ where r_1, r_2 are given by (11). Let (u, v) be a corresponding solution given by Theorem 1. Since $r_1 < qr_2$ and $r_2 < r_1 p$, from Theorem 1 we have that

$$(u, v) \in C([0, T_{\max}), L^{r_1}(\mathbb{R}^N) \cap C_0(\mathbb{R}^N)) \times C([0, T_{\max}), L^{r_2}(\mathbb{R}^N) \cap C_0(\mathbb{R}^N))$$

and (4) holds.

Let

$$\varphi(t) = \|u(t)\|_{r_1} + (t+1)^{\frac{N}{2}(\frac{1}{r_1} - \frac{1}{\eta_1})} \|u(t)\|_{\eta_1} + (t+1)^\alpha \|u(t)\|_\infty, \quad (40)$$

$$\psi(t) = \|v(t)\|_{r_2} + (t+1)^{\frac{N}{2}(\frac{1}{r_2} - \frac{1}{w_1})} \|v(t)\|_{w_1} + (t+1)^\beta \|v(t)\|_\infty \quad (41)$$

be functions defined for $t \in [0, T_{\max})$. We prove that there exists A_0 such that if $\varphi(0) + \psi(0) \leq A_0$ and $T \in (0, T_{\max})$, then φ, ψ are bounded on $[0, T]$, where $\alpha = \frac{p(1-\gamma_2)+(1-\gamma_1)}{pq-1}$, $\beta = \frac{q(1-\gamma_1)+(1-\gamma_2)}{pq-1}$ and $\eta_1, w_1 > 0$ are given by

$$\frac{1}{\eta_1} = \frac{1}{r_1} - \frac{2}{N}[\alpha + \mu]; \quad \frac{1}{w_1} = \frac{1}{r_2} - \frac{2}{N}(\beta + \lambda) \quad (42)$$

with $\mu, \lambda > 0$ satisfying

$$\frac{N}{2}\left(\frac{1}{r_1} - \frac{1}{q}\right) - \alpha < \mu < \min\left\{\gamma_1 - \alpha, \frac{N}{2r_1} - \alpha\right\}, \quad \mu < p\lambda, \quad (43)$$

$$\frac{N}{2}\left(\frac{1}{r_2} - \frac{1}{p}\right) - \beta < \lambda < \min\left\{\gamma_2 - \beta, \frac{N}{2r_2} - \beta\right\}, \quad \lambda < q\mu. \quad (44)$$

If this statement is proved, then $T^* = \infty$ and (u, v) is global.

From (42)–(44), we observe that

$$\frac{1}{r_2 q} < \frac{1}{\eta_1} < \frac{1}{q}, \quad \frac{1}{r_1 p} < \frac{1}{w_1} < \frac{1}{p} \quad (45)$$

and

$$\frac{1}{\eta_1} \leq \frac{p}{w_1}, \quad \frac{1}{w_1} \leq \frac{q}{\eta_1}. \quad (46)$$

Moreover, from (13), it easy to verify that

$$\begin{cases} \frac{q}{r_1} - \frac{2}{N} = \frac{2}{N} \frac{1 - q\gamma_1 + q(1 - p\gamma_2) + q(p+1)}{pq-1} < 1, \\ \frac{p}{r_2} - \frac{2}{N} = \frac{2}{N} \frac{1 - p\gamma_2 + p(1 - q\gamma_1) + p(q+1)}{pq-1} < 1, \end{cases} \quad (47)$$

$$\begin{cases} 1 - \gamma_1 + \alpha = \beta p, \\ 1 - \gamma_2 + \beta = \alpha q. \end{cases} \quad (48)$$

Estimate for $\|u\|_{r_1}$. From (48), we conclude that $\frac{1}{w_1} \leq \frac{1}{r_2} - \frac{2}{N}\beta < \frac{1}{p}(\frac{1}{r_1} + \frac{2}{N})$. From this fact, (47), (45) and (42) there exists w such that

$$\max\left\{\frac{1}{r_2} - \frac{2}{Np}, \frac{1}{w_1}, \frac{1}{r_1 p}\right\} < \frac{1}{w} < \min\left\{\frac{1}{p}, \frac{1}{p}\left(\frac{1}{r_1} + \frac{2}{N}\right), \frac{1}{r_2}\right\}.$$

Note that

$$0 \leq \frac{N}{2}\left(\frac{p}{w} - \frac{1}{r_1}\right) < 1, \quad \frac{N}{2}\left(\frac{p}{r_2} - \frac{p}{w}\right) < 1. \quad (49)$$

Since $w \in (r_2, w_1)$, it follows from interpolation inequality that

$$\begin{aligned} \|v\|_w &\leq \|v\|_{r_2}^\theta \|v\|_{w_1}^{1-\theta} \\ &\leq \|v\|_{r_2}^\theta \left[(t+1)^{\frac{N}{2}(\frac{1}{r_2} - \frac{1}{w_1})} \|v\|_{w_1}\right]^{1-\theta} (t+1)^{-\frac{N}{2}(\frac{1}{r_2} - \frac{1}{w})} \end{aligned} \quad (50)$$

where $\frac{1}{w} = \frac{\theta}{r_2} + \frac{1-\theta}{w_1}$.

Since $2 - \gamma_1 - \frac{N}{2}(\frac{p}{w} - \frac{1}{r_1}) - \frac{N}{2}(\frac{p}{r_2} - \frac{p}{w}) = 0$, by (2), (49) and (50), we obtain

$$\begin{aligned}
\|u(t)\|_{r_1} &\leq \|u_0\|_{r_1} + \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{w}-\frac{1}{r_1})} \int_0^s (s-\sigma)^{-\gamma_1} \|v(\sigma)\|_w^p d\sigma ds \\
&\leq \|u_0\|_{r_1} + \left[\sup_{s \in (0,t)} \psi(s) \right]^p \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{w}-\frac{1}{r_1})} \int_0^s (s-\sigma)^{-\gamma_1} \sigma^{-\frac{N}{2}(\frac{p}{r_2}-\frac{p}{w})} d\sigma ds \\
&\leq \|u_0\|_{r_1} + C \left[\sup_{s \in (0,t)} \psi(s) \right]^p.
\end{aligned}$$

Estimate for $\|u(t)\|_{\eta_1}$. As $\frac{p}{r_2} - \frac{4}{N} < \frac{1}{q}$ and $\frac{q}{r_1} - \frac{4}{N} < \frac{1}{p}$, exists w' satisfying

$$\max \left\{ \frac{1}{r_2} - \frac{2}{Np}, \frac{1}{\omega_1}, \frac{1}{p\eta_1} \right\} < \frac{1}{w'} \leq \min \left\{ \frac{1}{p}, \frac{1}{p} \left(\frac{1}{\eta_1} + \frac{2}{N} \right), \frac{1}{r_2} \right\}.$$

Note that $0 < \frac{N}{2}(\frac{p}{w'} - \frac{1}{\eta_1}) < 1$, $0 < \frac{N}{2}(\frac{p}{r_2} - \frac{p}{w'}) < 1$ and $2 - \gamma_1 + \frac{N}{2}(\frac{1}{r_1} - \frac{1}{\eta_1}) - \frac{N}{2}(\frac{p}{w'} - \frac{1}{\eta_1}) - \frac{N}{2}(\frac{p}{r_2} - \frac{p}{w'}) = 0$. From (50) (replacing w by w') and (37) we obtain

$$\begin{aligned}
\|u(t)\|_{\eta_1} &\leq (t+1)^{-\frac{N}{2}(\frac{1}{r_1}-\frac{1}{\eta_1})} (\|u_0\|_{r_1} + \|u_0\|_{\eta_1}) \\
&\quad + \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{w'}-\frac{1}{\eta_1})} \int_0^s (s-\sigma)^{-\gamma_1} \|v\|_{w'}^p d\sigma ds \\
&\leq (t+1)^{-\frac{N}{2}(\frac{1}{r_1}-\frac{1}{\eta_1})} (\|u_0\|_{r_1} + \|u_0\|_{\eta_1}) \\
&\quad + \left[\sup_{s \in (0,t)} \psi(s) \right]^p \int_0^t (t-s)^{-\frac{N}{2}(\frac{p}{w'}-\frac{1}{\eta_1})} \int_0^s (s-\sigma)^{-\gamma_1} \sigma^{-\frac{N}{2}(\frac{p}{r_2}-\frac{p}{w'})} d\sigma ds.
\end{aligned}$$

Thus, $(t+1)^{\frac{N}{2}(\frac{1}{r_1}-\frac{1}{\eta_1})} \|u(t)\|_{\eta_1} \leq (\|u_0\|_{r_1} + \|u_0\|_{\eta_1}) + C[\sup_{s \in (0,t)} \psi(s)]^p$.

Estimate for $\|u(t)\|_{\infty}$. We consider two situations.

Case a. Either $\frac{1}{p} \leq \frac{2}{Np}$ or $(\frac{p}{r_2} < \frac{4}{N}$ and $\frac{1}{w_1} < \frac{2}{Np})$. From (47), (45) and (42), there exists w'' such that

$$\max \left\{ \frac{1}{r_2} - \frac{2}{Np}, \frac{1}{w_1} \right\} < \frac{1}{w''} < \min \left\{ \frac{1}{p}, \frac{2}{Np}, \frac{1}{r_2} \right\}.$$

Note that $\frac{N}{2} \frac{p}{w''} < 1$ and $0 < \frac{Np}{2} (\frac{1}{r_2} - \frac{1}{w''}) < 1$. Since $w'' \in (r_2, w_1)$, using interpolation inequality as in (50), we obtain

$$\begin{aligned}
\|u(t)\|_{\infty} &\leq (t+1)^{-\frac{N}{2r_1}} (\|u_0\|_{r_1} + \|u_0\|_{\infty}) \\
&\quad + \int_0^t (t-s)^{-\frac{N}{2} \frac{p}{w''}} \int_0^s (s-\sigma)^{-\gamma_1} \|v(\sigma)\|_{w''}^p d\sigma ds \\
&\leq (t+1)^{-\frac{N}{2r_1}} (\|u_0\|_{r_1} + \|u_0\|_{\infty}) \\
&\quad + \left[\sup_{s \in (0,T)} \psi(s) \right]^p \int_0^t (t-s)^{-\frac{N}{2} \frac{p}{w''}} \int_0^s (s-\sigma)^{-\gamma_1} (\sigma+1)^{-\frac{Np}{2}(\frac{1}{r_2}-\frac{1}{w''})} d\sigma ds.
\end{aligned}$$

On the other hand, since $\alpha - \frac{N}{2r_1} = -\frac{p+1}{pq-1} < 0$, it follows that $\alpha + 2 - \gamma_1 - \frac{N}{2} \frac{p}{w''} - \frac{N}{2} (\frac{p}{r_2} - \frac{p}{w''}) < \frac{N}{2r_1} + 2 - \gamma_1 - \frac{N}{2} \frac{p}{r_2} = 0$. Therefore, from (39),

$$(t+1)^{\alpha} \|u(t)\|_{\infty} \leq (\|u_0\|_{r_1} + \|u_0\|_{\infty}) + C \left[\sup_{s \in (0,T)} \psi(s) \right]^p.$$

Case b. $\frac{1}{p} < \frac{2}{Np}$ and $(\frac{p}{r_2} \geq \frac{4}{N}$ or $\frac{2}{Np} \leq \frac{1}{w_1})$. From (47), (45) and (42), there exists w'' such that

$$\max \left\{ \frac{1}{r_2} - \frac{2}{Np}, \frac{1}{w_1}, \frac{2}{Np} \right\} < \frac{1}{w''} < \min \left\{ \frac{1}{p}, \frac{1}{r_2} \right\}. \quad (51)$$

Since $w'' \in (r_2, w_1)$ by interpolation inequality we conclude

$$\begin{aligned} \|u(t)\|_\infty &\leq (t+1)^{-\frac{N}{2r_1}} (\|u_0\|_{r_1} + \|u_0\|_\infty) \\ &\quad + \int_0^t (t+1-s)^{-\frac{N}{2} \frac{p}{w''}} \int_0^s (s-\sigma)^{-\gamma_1} (\|v\|_{w''}^p + \|v\|_\infty^p) d\sigma ds \\ &\leq (t+1)^{-\frac{N}{2r_1}} (\|u_0\|_{r_1} + \|u_0\|_\infty) + \left[\sup_{s \in (0,t)} \psi(s) \right]^p \\ &\quad \times \int_0^t (t+1-s)^{-\frac{N}{2} \frac{p}{w''}} \int_0^s (s-\sigma)^{-\gamma_1} [(\sigma+1)^{-\frac{Np}{2} (\frac{1}{r_2} - \frac{1}{w''})} + (\sigma+1)^{-\beta p}] d\sigma ds. \end{aligned}$$

Since $\beta p < 1$ we observe that $\frac{1}{r_2} - \frac{2}{Np} < \frac{1}{r_2} - \frac{2}{N}\beta$. From (42), $\frac{1}{w_1} < \frac{1}{r_2} - \frac{2}{N}\beta$. From (47) and (48), $\frac{2}{Np} < \frac{1}{r_2} - \frac{2}{N}\beta$. From these facts, we can assume that $\frac{1}{w''} \leq \frac{1}{r_2} - \frac{2}{N}\beta$. Thus, $\alpha + 1 - \gamma_1 - \frac{N}{2}(\frac{p}{r_2} - \frac{p}{w''}) = \beta p - \frac{N}{2}(\frac{p}{r_2} - \frac{p}{w''}) \leq 0$, and therefore we conclude as in the previous case.

From the previous estimates for $\|u\|_{r_1}$, $\|u\|_{\eta_1}$ and $\|u\|_\infty$ we obtain a constant C_1 (independent of t) such that

$$\varphi(t) \leq 4(\|u_0\|_{r_1} + \|u_0\|_\infty) + C_1 \left[\sup_{s \in (0,t)} \psi(s) \right]^p.$$

By similar computations it is possible to obtain $\psi(t) \leq 4(\|v_0\|_{r_2} + \|v_0\|_\infty) + C_2 [\sup_{s \in (0,t)} \varphi(s)]^q$. From here, doing $f(t) = \sup_{s \in (0,t)} \varphi(s)$ and $g(t) = \sup_{s \in (0,t)} \psi(s)$, we get

$$f(t) \leq 4(\|u_0\|_{r_1} + \|u_0\|_\infty) + C_1 g^p(t),$$

$$g(t) \leq 4(\|v_0\|_{r_2} + \|v_0\|_\infty) + C_2 f^q(t).$$

So,

$$\begin{aligned} f(t) &\leq 4(\|u_0\|_{r_1} + \|u_0\|_\infty) + C_1 [4(\|v_0\|_{r_2} + \|v_0\|_\infty) + C_2 f^q(t)]^p \\ &\leq 4(\|u_0\|_{r_1} + \|u_0\|_\infty) + 4C_1 2^{p-1} (\|v_0\|_{r_2} + \|v_0\|_\infty) + C_1 C_2^p 2^{p-1} f^{pq}(t) \\ &\leq A + B f^{pq}(t), \end{aligned}$$

with $A \geq 4(\|u_0\|_{r_1} + \|u_0\|_\infty) + 4C_1 2^{p-1} (\|v_0\|_{r_2} + \|v_0\|_\infty)$ and $B = C_1 C_2^p 2^{p-1}$. Let $h(x) = A + Bx^{pq} - x$, for $x \geq 0$. Then, $h(f(t)) \geq 0$. Note that $h(0) = A > 0$ and $h'(x_c) = 0$ if and only if $x_c = (pqB)^{-(pq-1)^{-1}}$. Moreover, $h(x_c) = A + (pqB)^{-1/(pq-1)} (\frac{1}{pq} - 1) = 0$ for $A = A_0 = (pqB)^{-1/(pq-1)} (1 - \frac{1}{pq})$. This implies, using the continuity of f , that if $4(\|u_0\|_{r_1} + \|u_0\|_\infty) + 4C_1 2^{p-1} (\|v_0\|_{r_2} + \|v_0\|_\infty) \leq A_0$, then $f(t) \leq x_c$.

The same argument can be used to show a bound for $g(t)$ assuming an appropriate bound for $4(\|v_0\|_{r_2} + \|v_0\|_\infty) + 4C_2 2^{q-1} (\|u_0\|_{r_1} + \|u_0\|_\infty)$.

4. Proof of the blow up part of Theorem 2

Lemma 5. Let $a, b, c, d > 0$, $p, q \geq 1$, $pq > 1$ and $(u, v) \in \{C^2([0, T])\}^2$ be a nontrivial solution of

$$\begin{cases} u'' + au' = bv^p, \\ v'' + cv' = du^q. \end{cases} \quad (52)$$

Given $T > 0$, there exists a constant $K = K(a, b, c, d, p, q, T) > 0$ so that if

$$u(0) \geq K \quad \text{or} \quad v(0) \geq K$$

and

$$u'(0) \geq -au(0) \quad \text{and} \quad v'(0) \geq -cv(0), \quad (53)$$

then there exists $T_{\max} < T$ so that (u, v) blows up in $T_{\max}$.

Remark 3. Lemma 5 extends to system (52) Lemma 2.1 of [2].

Proof of Lemma 5. We can assume that $u(0) > K$ and $v(0) > K$. In fact, if $u(0) > K$ and $v(0) = 0$, it follows from the continuity of u that there exists $T^* < T$ so that $u(t) > K$, for all $t \in [0, T^*]$. Integrating the second equation of (52) and using the condition (53), we obtain $v' + cv \geq d \int_0^t u^q(s) ds$. Hence, we can assume that $c > 1$ and $v' + cv \geq Mt$, for all $t \in [0, T^*]$ and some constant M . From here, we obtain $v(t) \geq \frac{M}{c}(t - \frac{1}{c} + e^{-ct})$, for all $t \in [0, T^*]$. It follows that $v(t) > 0$ for t sufficiently small.

The proof follows from the comparison principle, using a subsolution $(\bar{u}, \bar{v})$ of the system (52) of the kind $\bar{u}(t) = Ae^{-Bt}(T-t)^{-\alpha}$, $\bar{v}(t) = Ae^{-Bt}(T-t)^{-\beta}$, where A, B are appropriately chosen, $\alpha = 2\frac{p+1}{pq-1}$, and $\beta = 2\frac{q+1}{pq-1}$. $\square$

Proposition 6. Let $a, b, c, d, T > 0$, p, q and K as in Lemma 5. If $w_1, w_2 \in C^1([0, T])$, $(w_1(0), w_2(0)) \neq (0, 0)$, $w_1(0), w_2(0) \geq 0$ satisfying

$$\begin{cases} w_1' + aw_1 \geq b \int_0^t (t-s)^{-\gamma_1} w_2^p(s) ds, \\ w_2' + cw_2 \geq d \int_0^t (t-s)^{-\gamma_2} w_1^q(s) ds, \end{cases} \quad (54)$$

for all $t \in [0, T]$, then

- (i) If $T \geq 1$, then $w_1(0) < K$ and $w_2(0) < K$.
- (ii) If $T = \infty$, then $\sup_{t>0} w_1(t) \leq K$ and $\sup_{t>0} w_2(t) \leq K$.
- (iii) If $T = \infty$, then $\liminf_{t \rightarrow \infty} w_1(t) = 0$ and $\liminf_{t \rightarrow \infty} w_2(t) = 0$.
- (iv) If $T = \infty$, then $\liminf_{t \rightarrow \infty} t^{\gamma_1} w_1(t) > 0$ and $\liminf_{t \rightarrow \infty} t^{\gamma_2} w_2(t) > 0$.
- (v) If $T = \infty$, then

$$\liminf_{t \rightarrow \infty} t^{\alpha_1} w_1^{pq-1}(t) < \infty \quad \text{and} \quad \liminf_{t \rightarrow \infty} t^{\alpha_2} w_2^{pq-1}(t) < \infty,$$

where

$$\alpha_1 = p(1 - \gamma_2) + (1 - \gamma_1) \quad \text{and} \quad \alpha_2 = q(1 - \gamma_1) + (1 - \gamma_2). \quad (55)$$

- (vi) If $1 - p\gamma_2 + p(1 - q\gamma_1) \geq 0$ or $1 - q\gamma_1 + q(1 - p\gamma_2) \geq 0$, then $T < \infty$.

Remark 4.

- (i) Proposition 6 extends to system (54) the results of Proposition 2.1 of [2].
- (ii) In the proof of items (i) to (iii) we use the same arguments of [2]. Since we are dealing with a system, items (iv), (v) and (vi) are more delicate. In this cases, the arguments used in [2] do not follow directly.

Proof of Proposition 6. From (54), it follows that

$$\begin{cases} w_1(t) \geq e^{-at} w_1(0) + b \int_0^t \int_0^s e^{-a(t-s)} (s-\sigma)^{-\gamma_1} w_2^p(\sigma) d\sigma ds, \\ w_2(t) \geq e^{-ct} w_2(0) + d \int_0^t \int_0^s e^{-c(t-s)} (s-\sigma)^{-\gamma_2} w_1^q(\sigma) d\sigma ds \end{cases} \quad (56)$$

for all $t \in [0, T]$.

- (i) From (54), we have $w_1'(0) \geq -aw_1(0)$ and $w_2'(0) \geq -cw_2(0)$. Moreover, if $0 \leq t \leq 1$ then $w_1' + aw_1 \geq b \int_0^t w_2^p(s) ds$, $w_2' + cw_2 \geq d \int_0^t w_1^q(s) ds$. Therefore, $w_1 \geq u$ and $w_2 \geq v$ on $[0, 1]$, where (u, v) is solution of the system (52) with $u(0) = w_1(0)$, $v(0) = w_2(0)$, $u'(0) = -aw_1(0)$ and $v'(0) = -cw_2(0)$.

Since (w_1, w_2) is defined in $[0, 1]$, it follows from Lemma 5 that $w_1(0) < K$ and $w_2(0) < K$.

- (ii) Assume that $T = \infty$. Given $\tau \geq 0$, define $z_1(t) = w_1(t + \tau)$, $z_2(t) = w_2(t + \tau)$, for $t \geq 0$. We have

$$\begin{aligned}
z_1'(t) + az_1(t) &\geq b \int_0^{t+\tau} (\tau + t - s)^{-\gamma_1} w_2^p(s) ds \\
&\geq b \int_{\tau}^{t+\tau} (\tau + t - s)^{-\gamma_1} w_2^p(s) ds \\
&= b \int_0^t (t - s)^{-\gamma_1} z_2^p(s) ds.
\end{aligned}$$

Similarly, we get that $z_2'(t) + cz_2(t) \geq d \int_0^t (t - s)^{-\gamma_2} z_1^q(s) ds$.

It follows from (i) that $z_1(0) < K$ and $z_2(0) < K$, that is, $w_1(t) < K$ and $w_2(t) < K$, for all $t \geq 0$. Therefore, $\sup_{t \geq 0} w_1(t) \leq K$ and $\sup_{t \geq 0} w_2(t) \leq K$.

(iii) Let $T = \infty$. Assume that $w_1(t) \geq \eta > 0$, for all $t > 0$. It follows from (56) that $w_1(t) \geq \frac{b\eta_2^p e^{-at}}{1-\gamma_1} \int_0^t e^{as} s^{1-\gamma_1} ds$. For $t \geq 2$, we have

$$\begin{aligned}
w_1(t) &\geq \frac{b\eta_2^p e^{-at}}{1-\gamma_1} \int_{t-1}^t e^{as} s^{1-\gamma_1} ds \\
&\geq \delta t^{1-\gamma_1}
\end{aligned}$$

for some $\delta > 0$, which contradicts the property (ii). Therefore, $\liminf_{t \rightarrow \infty} w_1(t) = 0$. By a similar argument we conclude that $\liminf_{t \rightarrow \infty} w_2(t) = 0$.

(iv) We show that $\liminf_{t \rightarrow \infty} t^{\gamma_1} w_1(t) > 0$. Without loss of generality we assume that $w_1(0) > 0$. Indeed, since $(w_1, w_2) \neq 0$, we have that if $w_1(0) = 0$, then $w_2(0) > 0$. Replacing this inequality into the first inequality of (56) we obtain $w_1(t) \geq \frac{t^2}{2} e^{-(a+cp)t} w_2(0)^p > 0$, for all $t \in (0, 1)$.

From (56), $w_1(t) \geq w_1(0)e^{-at}$. Substituting this inequality into the second inequality of (56) we obtain $w_2(t) \geq Cw_1(0)^q t^{2-\gamma_2}$ for $t \in (0, 1)$ and some constant $C > 0$. Hence, for $t \geq 2$,

$$\begin{aligned}
w_1(t) &\geq b \int_{t-1}^t e^{-a(t-s)} \int_0^s (s-\sigma)^{-\gamma_1} w_2^p(\sigma) d\sigma ds \\
&\geq be^{-a} \int_{t-1}^t \int_0^1 (s-\sigma)^{-\gamma_1} w_2^p(\sigma) d\sigma ds \\
&\geq C \left[\inf_{1/2 \leq s < 1} w_2^p(s) \right] t^{-\gamma_1},
\end{aligned}$$

for some constant $C > 0$.

Analogously, we obtain that $\liminf_{t \rightarrow \infty} t^{\gamma_2} w_2(t) > 0$.

(v) Let $T = \infty$. It follows from property (iii) that there exists a sequence $\{s_n\}$ so that $w_2(s_n) = \min_{0 \leq s \leq s_n} w_2(s)$. In particular, we have $w_2'(s_n) \leq 0$. Note that

$$w_2'(s_n) \geq -cw_2(s_n) + \int_0^{s_n} (s_n - s)^{-\gamma_2} w_1^q(s) ds. \quad (57)$$

Moreover, if $n \in \mathbb{N}$ is such that $s_n \geq s$, we have

$$\begin{aligned}
w_1(s) &\geq b \int_0^s e^{-a(s-\eta)} \int_0^\eta (\eta - \sigma)^{-\gamma_1} w_2^p(\sigma) d\sigma d\eta \\
&\geq bw_2^p(s_n) \int_0^s e^{-a(s-\eta)} \int_0^\eta (\eta - \sigma)^{-\gamma_1} d\sigma d\eta.
\end{aligned} \quad (58)$$

Since $w_2'(s_n) \leq 0$, it follows from (57), (58) that

$$\begin{aligned}
c w_2(s_n) &\geq C \int_0^{s_n} (s_n - s)^{-\gamma_2} w_2^{pq}(s_n) \left[\int_0^s e^{-a(s-\eta)} \int_0^\eta (\eta - \sigma)^{-\gamma_1} d\sigma d\eta \right]^q ds \\
&\geq C \int_1^{s_n} (s_n - s)^{-\gamma_2} w_2^{pq}(s_n) \left[\int_0^s e^{-a(s-\eta)} \eta^{1-\gamma_1} d\eta \right]^q ds,
\end{aligned} \tag{59}$$

where $C > 0$ is a constant. Note that

$$\int_0^s e^{-a(s-\eta)} \eta^{1-\gamma_1} d\eta \geq \int_{s-1}^s e^{-a(s-\eta)} \eta^{1-\gamma_1} d\eta \geq C(s-1)^{1-\gamma_1},$$

for $s \geq 1$, so that by (59)

$$C \geq w_2^{pq-1}(s_n) \int_1^{s_n} (s_n - s)^{-\gamma_2} (s-1)^{(1-\gamma_1)q} ds = w_2^{pq-1}(s_n) (s_n - 1)^{(1-\gamma_2)+(1-\gamma_1)q}.$$

Therefore, the result for w_2 follows. Similarly, we obtain the result for w_1 .

(vi) Assume that $1 - p\gamma_2 + p(1 - q\gamma_1) \geq 0$. We consider two cases.

Case 1. $1 - p\gamma_2 + p(1 - q\gamma_1) > 0$. From (iv), it follows that $t^{\gamma_1} w_1(t) > \delta$, for some $\delta > 0$. Hence, we have

$$t^{\alpha_1} w_1^{pq-1}(t) > C t^{\alpha_1 - \gamma_1(pq-1)} = C t^{p(1-\gamma_2) + pq(1-\gamma_1) - pq + 1}.$$

From (v), we have $T < \infty$.

Case 2. $1 - p\gamma_2 + p(1 - q\gamma_1) = 0$. From (54), $w_1' + a w_1 \geq 0$. Using this fact and (iv), it follows that $(1+t)^{\gamma_1} w_1(t) > \delta$ for all $t \geq 0$ and some $\delta > 0$. Hence, by (56) we have that

$$\begin{aligned}
w_2(t) &\geq C \int_{t-1}^t e^{-a(t-s)} \int_0^s (s-\sigma)^{-\gamma_2} (1+\sigma)^{-\gamma_1 q} d\sigma ds \\
&\geq C(t+1)^{-\gamma_1 q} (t-1)^{1-\gamma_2}.
\end{aligned}$$

Using (56) again, for $t \geq 3$, we obtain

$$\begin{aligned}
w_1(t) &\geq C \int_{t-1}^t e^{-a(t-s)} \int_2^s (s-\sigma)^{-\gamma_1} (1+\sigma)^{-\gamma_1 pq} (\sigma-1)^{(1-\gamma_2)p} d\sigma ds \\
&\geq C \int_{t-1}^t e^{-a(t-s)} \int_2^s (s-\sigma)^{-\gamma_1} (1+\sigma)^{-\gamma_1 pq + (1-\gamma_2)p} d\sigma ds.
\end{aligned} \tag{60}$$

Since $(1-\gamma_2)p - pq\gamma_1 = -1$, it follows from (60) that $w_1(t) \geq C t^{-\gamma_1} \ln t$, for t sufficiently large. Therefore,

$$t^{\alpha_1} w_1^{pq-1}(t) \geq C t^{\alpha_1 - \gamma_1(pq-1)} (\ln t)^{pq-1} = C (\ln t)^{pq-1}.$$

Thus, $t^{\alpha_1} w_1^{pq-1}(t) \rightarrow \infty$ as $t \rightarrow \infty$, which contradicts the estimate (v). $\square$

Proof of blow up part of Theorem 2. Let (u, v) be a nonnegative solution of system (1) with $(u_0, v_0) \neq 0$, $u_0, v_0 \geq 0$.

Firstly, suppose that (17) is valid and, without loss of generality, assume that $1 - p\gamma_2 + p(1 - q\gamma_1) \geq 0$. Moreover, assume that the solution (u, v) exists globally.

Let $\chi(x) = c e^{-\sqrt{N^2 + |x|^2}}$, where c is such that $\|\chi\|_{L^1} = 1$. We have that $\Delta \chi \geq -\chi$. Let $f_1(t) = \int_{\mathbb{R}^N} u(t, x) \chi dx$, $f_2(t) = \int_{\mathbb{R}^N} v(t, x) \chi dx$. It follows, from (1) and Jensen's inequality,

$$\begin{cases} f_1' + f_1 \geq \int_0^t (t-s)^{-\gamma_1} f_2^p(s) ds, \\ f_2' + f_2 \geq \int_0^t (t-s)^{-\gamma_2} f_1^q(s) ds, \end{cases}$$

which contradicts the property (vi) of the previous proposition.

Suppose now that (16) holds. Without loss of generality, assume that $1 - p\gamma_2 + p(1 - q\gamma_1) + p(q + 1) \geq \frac{N}{2}(pq - 1)$. We divide the analysis into two cases.

Case I. $1 - p\gamma_2 + p(1 - q\gamma_1) + p(q + 1) > \frac{N}{2}(pq - 1)$. Assume that the solution (u, v) exists globally and consider $u(t, x) = t^{\beta_1} U(s, y)$, $v(t, x) = t^{\beta_2} V(s, y)$, where $\beta_1 = -\frac{(2-\gamma_1)+p(2-\gamma_2)}{pq-1}$, $\beta_2 = -\frac{(2-\gamma_2)+q(2-\gamma_1)}{pq-1}$, $s = \ln t$ and $y = \frac{x}{\sqrt{t}}$. Note that (U, V) is a solution of the system

$$\begin{cases} U_s + LU + \beta_1 U = \int_0^1 (1-\eta)^{-\gamma_1} \eta^{p\beta_2} V^p \left(s + \ln \eta, \frac{y}{\sqrt{\eta}} \right) d\eta, \\ V_s + LV + \beta_2 V = \int_0^1 (1-\eta)^{-\gamma_2} \eta^{q\beta_1} U^q \left(s + \ln \eta, \frac{y}{\sqrt{\eta}} \right) d\eta, \end{cases} \quad (61)$$

where $L = -\Delta - \frac{1}{2}y \cdot \nabla$. Let $\varphi(y) = e^{-\frac{|y|^2}{4}}$. Note that $L\varphi = \frac{N}{2}\varphi$. Moreover, if $K(y) = e^{\frac{|y|^2}{4}}$, then $\int_{\mathbb{R}^N} (Lf)gK = \int_{\mathbb{R}^N} f(Lg)K$ as long as f, g are sufficiently smooth functions. Let $\Psi = (4\pi)^{-\frac{N}{2}}\varphi^2$. Note that $\int_{\mathbb{R}^N} \Psi K = 1$. Moreover, we can easily check that $L\Psi \geq N\Psi$. Now, multiplying the first equation of (61) by ΨK and integrating in y , we have

$$\begin{aligned} \frac{d}{ds} \int_{\mathbb{R}^N} U \Psi K dy + (N + \beta_1) \int_{\mathbb{R}^N} U \Psi K dy \\ \geq \int_0^1 (1-\eta)^{-\gamma_1} \eta^{p\beta_2} \int_{\mathbb{R}^N} V^p \left(s + \ln \eta, \frac{y}{\sqrt{\eta}} \right) \Psi K dy d\eta. \end{aligned} \quad (62)$$

Note that $\int_{\mathbb{R}^N} V^p(s + \ln \eta, \frac{y}{\sqrt{\eta}}) \Psi K dy = \eta^{\frac{N}{2}} \int_{\mathbb{R}^N} V^p(s + \ln \eta, y) [\Psi K](y\sqrt{\eta}) dy$. Since $\eta \leq 1$, we have $[\Psi K](y\sqrt{\eta}) \geq [\Psi K](y)$. It follows from Jensen's inequality that

$$\int_{\mathbb{R}^N} V^p \left(s + \ln \eta, \frac{y}{\sqrt{\eta}} \right) \Psi K dy \geq \eta^{\frac{N}{2}} \left[\int_{\mathbb{R}^N} V(s + \ln \eta, y) [\Psi K](y) dy \right]^p. \quad (63)$$

Let $w_1(s) = \int_{\mathbb{R}^N} U(s) \Psi K dy$, $w_2(s) = \int_{\mathbb{R}^N} V(s) \Psi K dy$, $a = \max\{N + \beta_1, 0\}$ and $c = \max\{N + \beta_2, 0\}$. From (62) and (63), it follows that

$$\begin{aligned} w'_1 + aw_1 &\geq \int_0^1 (1-\eta)^{-\gamma_1} \eta^{\frac{N}{2}-p\beta_2} w_2^p(s + \ln \eta) d\eta \\ &\geq \min_{\frac{1}{e} \leq \eta \leq 1} \eta^{\frac{N}{2}-p\beta_2} \int_{\frac{1}{e}}^1 w_2^p(s + \ln \eta) d\eta. \end{aligned}$$

Thus, there exists a constant $b > 0$ such that $w'_1 + aw_1 \geq b \int_{s-1}^s w_2^p(\sigma) d\sigma$. Analogously, by similar calculations, we obtain $w'_2 + cw_2 \geq d \int_{s-1}^s w_1^q(\sigma) d\sigma$ for some constant $d > 0$. Define now $z_1(s) = w_1(t + s)$ and $z_2(s) = w_2(t + s)$ for $t \geq 0$. Then for all $0 \leq s \leq 1$,

$$\begin{cases} z'_1 + az_1 \geq b \int_{s-1}^s z_2^p(\sigma) d\sigma \geq b \int_0^s z_2^p(\sigma) d\sigma, \\ z'_2 + cz_2 \geq d \int_{s-1}^s z_1^q(\sigma) d\sigma \geq d \int_0^s z_1^q(\sigma) d\sigma. \end{cases}$$

The same argument used in the proof of Proposition 6 shows that there exists a constant $K > 0$ (independent of t) so that $z_1(0), z_2(0) < K$. Hence, we conclude that

$$\sup_{s \geq 0} w_1(s), \sup_{s \geq 0} w_2(s) < \infty. \quad (64)$$

Consider the heat kernel $G(t, x) = (4\pi t)^{-\frac{N}{2}} e^{-\frac{|x|^2}{4t}}$. Note that, for $\tau, p > 0$, we have

$$G(\tau)^p = p^{-\frac{N}{2}} (4\pi \tau)^{-\frac{N}{2}(p-1)} G\left(\frac{\tau}{p}\right). \quad (65)$$

It follows from the system (1) that $u(t) \geq G(t) * u_0$ and $v(t) \geq G(t) * v_0$. Since u_0 and v_0 are nonnegative and $u_0, v_0 \neq 0$, we have $u(1) \geq \epsilon G(\frac{1}{2})$, $v(1) \geq \tilde{\epsilon} G(\frac{1}{2})$ where $\epsilon, \tilde{\epsilon} > 0$.

From (3), it follows that $u(t) \geq \epsilon G(t - \frac{1}{2})$, $v(t) \geq \tilde{\epsilon} G(t - \frac{1}{2})$ for $t \geq 1$. Hence, if $\epsilon, \tilde{\epsilon}$ are sufficiently small, then

$$u(t) \geq \epsilon G\left(\frac{t}{2}\right), \quad v(t) \geq \tilde{\epsilon} G\left(\frac{t}{2}\right) \quad (66)$$

for $t \geq 1$. From (2), (66) and (65), it follows that for $t \geq 2$

$$\begin{aligned} u(t) &\geq \int_1^t \int_1^s S(t-s)(s-\sigma)^{-\gamma_1} v^p(\sigma) d\sigma ds \\ &\geq C \int_1^t \int_1^s S(t-s)(s-\sigma)^{-\gamma_1} G^p\left(\frac{\sigma}{2}\right) d\sigma ds \\ &\geq C \int_1^t \int_1^s S(t-s)(s-\sigma)^{-\gamma_1} \sigma^{-\frac{N(p-1)}{2}} G\left(\frac{\sigma}{2p}\right) d\sigma ds \\ &\geq C \int_1^t \int_1^s (s-\sigma)^{-\gamma_1} \sigma^{-\frac{N(p-1)}{2}} G\left(t-s+\frac{\sigma}{2p}\right) d\sigma ds, \end{aligned} \quad (67)$$

where C is a constant. Note that, if $1 \leq \sigma \leq s \leq \frac{t}{2}$, then $\frac{t}{2} \leq t-s+\frac{\sigma}{2p} \leq t$. Therefore,

$$G\left(t-s+\frac{\sigma}{2p}\right) \geq 2^{-\frac{N}{2}} G\left(\frac{t}{2}\right). \quad (68)$$

From (67) and (68), we have

$$\begin{aligned} u(t) &\geq CG\left(\frac{t}{2}\right) \int_1^{\frac{t}{2}} \int_1^s (s-\sigma)^{-\gamma_1} \sigma^{-\frac{N(p-1)}{2}} d\sigma ds \\ &\geq CG\left(\frac{t}{2}\right) \int_1^{\frac{t}{2}} \left(\frac{t}{2}-\sigma\right)^{1-\gamma_1} \sigma^{-\frac{N(p-1)}{2}} d\sigma \\ &\geq CG\left(\frac{t}{2}\right) t^{1-\gamma_1} \end{aligned}$$

for t sufficiently large. If we use (66) for t small, we conclude that, for $t \geq 1$

$$u(t) \geq Ct^{1-\gamma_1} G\left(\frac{t}{2}\right) = Ct^{1-\gamma_1-\frac{N}{2}} e^{-\frac{|x|^2}{4t}}. \quad (69)$$

Hence, $w_1(s) \geq Ct^{1-\gamma_1-\frac{N}{2}-\beta_1}$. By assumption, we have $1-\gamma_1-\frac{N}{2}-\beta_1 > 0$, which contradicts (64).

Case II. $1-p\gamma_2+p(1-q\gamma_1)+p(q+1)=\frac{N}{2}(pq-1)$. Using the estimates (69) and (68), we obtain, for $t \geq 4$,

$$\begin{aligned} v(t) &\geq \int_1^t \int_1^s S(t-s)(s-\sigma)^{-\gamma_2} u^q(\sigma) d\sigma ds \\ &\geq C \int_1^t \int_1^s S(t-s)(s-\sigma)^{-\gamma_2} \sigma^{q(1-\gamma_1)} G\left(\frac{\sigma}{2}\right)^q d\sigma ds \end{aligned}$$

$$\begin{aligned}
&\geq C \int_1^t \int_1^s (s-\sigma)^{-\gamma_2} \sigma^{q(1-\gamma_1)-\frac{N}{2}(q-1)} G\left(t-s+\frac{\sigma}{2q}\right) d\sigma ds \\
&\geq CG\left(\frac{t}{2}\right) \int_1^{\frac{t}{2}} \int_1^s (s-\sigma)^{-\gamma_2} \sigma^{q(1-\gamma_1)-\frac{N}{2}(q-1)} d\sigma ds \\
&\geq CG\left(\frac{t}{2}\right) \int_1^{\frac{t}{2}} s^{-\gamma_2} \left(\int_1^s \sigma^{q(1-\gamma_1)-\frac{N}{2}(q-1)} d\sigma \right) ds \\
&= CG\left(\frac{t}{2}\right) \int_1^{\frac{t}{2}} s^{q(1-\gamma_1)-\frac{N}{2}(q-1)+1-\gamma_2} ds \\
&= CG\left(\frac{t}{2}\right) t^{q(1-\gamma_1)-\frac{N}{2}(q-1)+2-\gamma_2}.
\end{aligned}$$

From the previous estimate for v , it follows that, for t sufficiently large,

$$\begin{aligned}
u(t) &\geq \int_4^t \int_4^s S(t-s)(s-\sigma)^{-\gamma_1} v^p(\sigma) d\sigma ds \\
&\geq C \int_4^t \int_4^s S(t-s)(s-\sigma)^{-\gamma_1} \sigma^{pq(1-\gamma_1)-\frac{Np(q-1)}{2}+p(2-\gamma_2)} G\left(\frac{\sigma}{2}\right)^p d\sigma ds \\
&\geq C \int_4^{\frac{t}{2}} \int_4^s (s-\sigma)^{-\gamma_1} \sigma^{p(2-\gamma_2)+pq(1-\gamma_1)-\frac{N(pq-1)}{2}} G\left(t-s+\frac{\sigma}{2p}\right) d\sigma ds \\
&\geq CG\left(\frac{t}{2}\right) \int_4^{\frac{t}{2}} \int_4^s (s-\sigma)^{-\gamma_1} \sigma^{-1} d\sigma ds \\
&\geq CG\left(\frac{t}{2}\right) t^{1-\gamma_1} \ln(t/4).
\end{aligned}$$

Hence, if $w_1(\sigma) = \int_{\mathbb{R}^N} U(\sigma) \Psi K dy$, then

$$\begin{aligned}
w_1(\sigma) &= \int_{\mathbb{R}^N} \sigma^{-\beta_1} u(\sigma) \Psi K dy \\
&\geq C \int_{\mathbb{R}^N} \sigma^{-\beta_1+(1-\gamma_1)} \ln(\sigma/4) G_{\frac{\sigma}{2}} \Psi K dy \\
&= C \int_{\mathbb{R}^N} \sigma^{-\beta_1+(1-\gamma_1)-\frac{N}{2}} \ln(\sigma/4) e^{-\frac{|y|^2}{2\sigma}} \Psi K dy \\
&= C \ln(\sigma/4),
\end{aligned}$$

which contradicts (64) for σ large. $\square$

Remark 5. To conclude the estimates (64) and (69), we use similar arguments to those used in [2]. In the proof of the case II, the arguments used here not follow directly of [2].

5. Bounded domains

We remind that the existence of a local solution for the system (18) is guaranteed by Remark 2. Here $\{S(t)\}_{t \geq 0}$ denotes the heat semigroup in Ω , with Dirichlet boundary conditions. We denote by $\lambda_1 > 0$ the first eigenvalue of $-\Delta$ in $H_0^1(\Omega)$ and by φ_1 the corresponding eigenvector satisfying $\int_{\Omega} \varphi_1 = 1$.

Proof of Theorem 3. (i) Assume that condition (19) holds. The proof is standard and is based in the fixed point argument. Let $\alpha = \frac{(1-\gamma_2)+q(1-\gamma_1)}{pq-1}$, $\beta = \frac{(1-\gamma_1)+p(1-\gamma_2)}{pq-1}$. Note that $\alpha, \beta > 0$ are such that $\alpha p, \beta q < 1$ and $1-\gamma_1+\beta = \alpha p$, $1-\gamma_2+\alpha = \beta q$.

Let $E = \{L^\infty((0, \infty), L^\infty(\Omega))\}^2$ and $K = \{(u, v) \in E; (1+t)^\alpha \|v(t)\|_\infty \leq \delta_1, (1+t)^\beta \|u(t)\|_\infty \leq \delta_2 \text{ for } t \geq 0\}$, where $\delta_1, \delta_2 > 0$ are small and conveniently chosen. The space K with the metric

$$d((u, v), (\bar{u}, \bar{v})) = \sup_{t \geq 0} [(1+t)^\alpha \|u - \bar{u}\|_\infty + (1+t)^\beta \|v - \bar{v}\|_\infty]$$

is a complete metric space. Since Ω is bounded, we deduce that

$$\begin{aligned} (1+t)^\alpha \|S(t)u_0\|_\infty &\leq C(1+t)^\alpha e^{-t\lambda_1} \|u_0\|_\infty \leq A \|u_0\|_\infty, \\ (1+t)^\beta \|S(t)v_0\|_\infty &\leq C(1+t)^\beta e^{-t\lambda_1} \|v_0\|_\infty \leq B \|v_0\|_\infty, \end{aligned} \quad (70)$$

where A and B are independent of u_0 and v_0 .

Consider the map $\psi : K \rightarrow E$ given by (23). Given $(u, v) \in E$, we have

$$\begin{aligned} \|\varphi_1(v)(t) - S(t)u_0\|_\infty &\leq \delta_2^p \int_0^t \int_0^s e^{-\lambda_1(t-s)} (s-\sigma)^{-\gamma_1} (1+\sigma)^{-p\alpha} d\sigma ds \\ &\leq \delta_2^p \int_0^t \int_0^s e^{-\lambda_1(t-s)} (s-\sigma)^{-\gamma_1} \sigma^{-p\alpha} d\sigma ds \\ &= C \delta_2^p \int_0^t e^{-\lambda_1(t-s)} s^{1-\gamma_1-p\alpha} ds \\ &= C \delta_2^p \int_0^t e^{-\lambda_1(t-s)} s^{-\beta} ds \\ &\leq C \delta_2^p (1+t)^{-\beta}, \end{aligned} \quad (71)$$

where the last inequality follows from l'Hôpital's rule.

Analogously, we obtain $\|\varphi_2(u)(t) - S(t)v_0\|_\infty \leq C \delta_1^q (1+t)^{-\alpha}$. A similar estimate can be obtained for the differences $(\varphi_1(v) - \varphi_1(\bar{v}))$ and $(\varphi_2(u) - \varphi_2(\bar{u}))$. Therefore, we conclude that $\psi : K \rightarrow K$ has a unique fixed point (u, v) , which is a solution of the system (18).

(ii) Assume that (20) holds. Let $f_1(t) = \int_\Omega v(t)\varphi_1$ and $f_2(t) = \int_\Omega u(t)\varphi_1$. Multiplying each equation of the system (18) by φ_1 , integrating over Ω , and using Jensen's inequality, we obtain

$$\begin{cases} f_1'(t) + \lambda_1 f_1(t) \geq \int_0^t (t-s)^{-\gamma_1} f_2^p(s) ds, \\ f_2'(t) + \lambda_1 f_2(t) \geq \int_0^t (t-s)^{-\gamma_2} f_1^q(s) ds. \end{cases} \quad (72)$$

The result follows from Proposition 6, item (vi). $\square$

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